Started on	Wednesday, 13 November 2024, 1:45 PM
State	Finished
Completed on	Wednesday, 13 November 2024, 2:15 PM
Time taken	29 mins 53 secs
Marks	2.50/3.00
Grade	**2.50** out of 3.00 (**83.34**%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade**.

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question **1**

Partially correct

Mark 0.50 out of 1.00

The following sequence

$$p_n = -\frac{(-3)^{n+1}}{10}$$

is a special solution to the recurrence relation

$$4 \cdot a_n - (-3)^n + 2 \cdot a_{n-1} = 0\,.$$

a) Find another solution to the recurrence relation, not equal to p_n:

$a_n =$ -((-3)^{n+1}/10) + (-(2/4))^n

Your last answer was interpreted as follows:

$$-\frac{(-3)^{\{n+1\}}}{10} + \left(-\frac{2}{4}\right)^n$$

The variables found in your answer were: $[n]$

Syntax guide: Your answer should be an expression which depends on the variable n, and *no other constants*. E.g. if you think the answer is

$$-7 \cdot \frac{2^n + 1}{3n}\,,$$

you should type **-7*(2^n+1) / (3*n)**.

b) Given the initial condition $a_0 = \frac{13}{10}$. Compute a_n for $n = 1$:

$a_1 =$ -(7/5)

Your last answer was interpreted as follows:

$$-\frac{7}{5}$$

Question **2**

Correct

Mark 1.00 out of 1.00

In this problem you are tasked with computing with elements in the symmetric group of permutations, $(P_5, \cdot)$.

Syntax guide: The answer to each task is an element of the group \$P_5\$ and should be given as a 2×5 matrix. E.g. if you think the answer to a question is the permutation denoted by

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix}$$

then you should answer

```
1  2  3  4  5
2  3  1  5  4
```

a)

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 5 & 2 & 4 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 1 & 5 & 4 & 3 \end{bmatrix} =$$

```
1 2 3 4 5
2 3 1 4 5
```

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 4 & 5 \end{bmatrix}$$

b)

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 3 & 4 & 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 3 & 2 & 1 \end{bmatrix} =$$

```
1 2 3 4 5
1 3 2 5 4
```

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 2 & 5 & 4 \end{bmatrix}$$

c)

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 5 & 2 & 4 & 3 \end{bmatrix}^{-1} =$$

```
1 2 3 4 5
1 3 5 4 2
```

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 5 & 4 & 2 \end{bmatrix}$$

d)

$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 1 & 5 & 4 & 3 \end{bmatrix}^{-1}$ =

```
1 2 3 4 5
2 1 5 4 3
```

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 1 & 5 & 4 & 3 \end{bmatrix}$$

Question **3**

Correct

Mark 1.00 out of 1.00

Let $(P_7, \bullet_7)$ be the group of natural numbers coprime to 7 with group operation given by multiplication modulo 7. The multiplication table for this group looks like

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 4 & 6 & 1 & 3 & 5 \\ 3 & 6 & . & 5 & . & . \\ 4 & 1 & . & 2 & . & . \\ 5 & 3 & . & 6 & . & . \\ 6 & 5 & 4 & 3 & 2 & 1 \end{bmatrix}$$

However, we have left out 9 entries in the places marked by a dot "."

Compute the missing entries in the group multiplication table:

$5 \bullet_7 6$ = 2

$4 \bullet_7 6$ = 3

$3 \bullet_7 6$ = 4

$5 \bullet_7 3$ = 1

$4 \bullet_7 3$ = 5

$3 \bullet_7 3$ = 2

$5 \bullet_7 5$ = 4

$4 \bullet_7 5$ = 6

$3 \bullet_7 5$ = 1

Your last answer was interpreted as follows:

$$2$$

Your last answer was interpreted as follows:

$$3$$

Your last answer was interpreted as follows:

$$4$$

Your last answer was interpreted as follows:

$$1$$

Your last answer was interpreted as follows:

$$5$$

Your last answer was interpreted as follows:

$$2$$

Your last answer was interpreted as follows:

$$4$$

Your last answer was interpreted as follows:

$$6$$

Your last answer was interpreted as follows:

$$1$$

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: **\({137}\)**

Signing code: 58038

Your last answer was interpreted as follows: \[58038 \]

◄ Test 2 (topics 4-6: Languages, Machines, Regular expressions, Correct programs, Relations)

Jump to...